

# extracting elemental iodine from an amazon product [WIP]

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## lugol

hi! I've been wanting to make a relatively harmless and interesting extraction in my homelab, and I've realized Lugol (a solution containing iodine and potassium iodide dissolved in water) might be a good starting material. my objective is to separate the elemental iodine from the other components on Lugol, and ultimately obtain relatively pure elemental iodine ( $I_2$ ) crystals at the end of my experiment.

## starting material

Lugol's solution is an aqueous mixture that contains elemental iodine ( $I_2$ ) and potassium iodide (KI). the iodine isn't present exclusively as isolated  $I_2$  molecules in the aqueous solution. rather, iodide ions ( $I^-$ ) supplied by the potassium iodide (KI) interact with molecular iodine ( $I_2$ ) to form  $I_3^-$ , called triiodide:  **$I_2 + I^- \rightleftharpoons I_3^-$**

## why does it happen?

when **KI (potassium iodide)** dissolves in water, it separates into potassium ions ( $K^+$ ) and iodide ions ( $I^-$ ):  **$KI \rightarrow K^+ + I^-$**

the iodide ion can interact with an iodine molecule afterwards.  $I_2$  has an electron-rich distribution, and  $I^-$  can interact with it to form the triiodide ion,  $I_3^-$ . the important consequence is that  **$I_3^-$  is much more comfortable existing in water than  $I_2$  is**. because  $I_3^-$  is a charged species, water can stabilize it much more effectively than it can neutral  $I_2$ , because water is a polar molecule, as you may know. the iodide supplied by KI gives the iodine another, much more water-soluble form to exist as. this allows considerably more iodine to remain dissolved in the aqueous solution.